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REAL OPTIONS IN STRATEGIC MANAGEMENT

Tony W. Tong and Jeffrey J. Reuer

A fundamental issue in the field of strategic management concerns firms' strategic choices and directions (Rumelt, Schendel, & Teece, 1994). Reflecting this central concern, a substantial amount of research in the field has examined the antecedents of a wide range of strategic decisions by firms as well as their performance implications. Whether strategic decisions involve internal investments in technology or external corporate development activities, they generally involve resource commitments to future initiatives under uncertainty. As a result, the role of uncertainty has received a great deal of attention in strategy research, and there has been recurrent interest in how firms might better manage strategic decision making under uncertainty.

Research has long recognized the key role that uncertainty plays in organizations and management (e.g., Cyert & March, 1963; Thompson, 1967), yet a recent and novel treatment of uncertainty comes from real options theory. In contrast to traditional views that managerial discretion is limited in the face of uncertainty or that organizational inertia dominates, real options theory maintains that firms can engage uncertainty and benefit by investing in options to respond to uncertain futures and by managing the investments in a sequential fashion as uncertainty is resolved (Kogut, 1991; Dixit & Pindyck, 1994; Kogut & Kulatilaka, 2001). Recent advances in strategy and finance have suggested that real options theory potentially offers a powerful valuation tool as well as a systematic strategy framework

Real Options Theory

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to evaluate and structure resource investments under uncertainty, and that successful use of real options can lead to the benefits of downside risk reduction and upside potential enhancement (Bowman & Hurry, 1993; Kogut & Kulatilaka, 1994a; Trigeorgis, 1996; McGrath, 1997; Amram & Kulatilaka, 1999).

In undertaking this volume, our objectives are two-fold. First, as interest in real options theory continues to grow, there have also been questions on the greater promise of real options theory in strategy. While advocates believe that real options theory informs strategic decision making under uncertainty, others also see difficulties surrounding the theory's larger applicability to strategic management issues. We suggest that part of this dialogue reflects broader questions on how real options theory might link to the foundations of the strategy field, and we identify four fundamental questions for real options theory to advance in strategy. Second, the strategy literature on real options has developed rapidly, and research has examined diverse aspects of the theory. As such, our second objective is to catalog, synthesize, and critique the extant real options research in strategy. This effort can delineate the ways in which real options theory contributes to strategy, and it also can reveal certain avenues for future research on real options. The focused volume therefore can provide a forum for researchers to tackle key questions, discuss promising opportunities, and map out the future research agenda for real options theory in strategic management.

In the following section, we briefly review the origins of real options theory, trace its developments in strategic management, and outline three reasons why it has become important for the field. This review and assessment leads to an overarching framework that we also use to organize the remaining 17 chapters in this volume, and we highlight how these articles are built on the framework and contribute to our expanded knowledge. We conclude by offering four fundamental questions that we believe lie at the interface between real options and strategy and can help move forward real options research in strategy in important ways.

THE DEVELOPMENT OF REAL OPTIONS THEORY

The Origins of Real Options Theory

Real options theory begins by drawing an analogy between real options and financial options. A financial option is a derivative security whose value is

derived from the worth and characteristics of another financial security, or the so-called underlying asset. By definition, a financial option gives its holder the right, but not the obligation, to buy or sell the underlying asset at a specified price (i.e., the exercise price) on or before a given date (i.e., the expiration date). Financial economists Black and Scholes (1973) and Merton (1973) pioneered a formula for the valuation of a financial option, and their methodology has opened up the subsequent research on the pricing of financial assets and paved the way for the development of real options theory.

The notion of real options was developed from Myers' (1977) seminal idea that one can view firms' discretionary investment opportunities as a call option on real assets, in much the same way as a financial call option provides decision rights on financial assets. By way of analogy, a real option has as its underlying asset the gross project value of expected operating cash flows; its exercise price is the investment required to obtain this underlying asset; and the time to maturity is the period of time during which the decision maker can defer the investment before the investment opportunity expires (e.g., Myers, 1977; Trigeorgis, 1996). Formally stated, real options are investments in real assets, as opposed to financial assets, which confer the firm the right, but not the obligation, to undertake certain actions in the future (e.g., Trigeorgis, 1996; Amram & Kulatilaka, 1999). Comparisons of financial and real options can be found in standard textbooks (e.g., Brealey, Myers, & Allen, 2006).

Real options research in finance and economics has developed a taxonomy of common real options that are often embedded in an investment, including deferral options, options to stage investments, options to alter operating scale, abandonment options, switching options, and growth options. In addition, an investment frequently involves a combination of some of the common real options above, and their combined value often differs from the sum of the value of each option in isolation (Trigeorgis, 1993). Investments such as technology development or venture capital also may consist of sequential stages, and such multistage investments comprise compound options, whose underlying asset is not a real asset, but another option (Roberts & Weitzman, 1981; Trigeorgis, 1996). To the extent that an investor can hold a portfolio of options simultaneously (Merton, 1973), a firm undertaking multiple investments at a point in time may also experience option portfolio interactions, in that options embedded in one investment may shape the value of other options held by the firm and therefore the overall value of the option portfolio (e.g., Triantis & Hodder, 1990; Luehrman, 1998; Smit & Trigeorgis, 2004).

The real options literature in finance and economics tends to have an analytic focus, employing real options analysis to evaluate firms' investments under uncertainty and to model the optimal conditions for undertaking such investments. For example, earlier research in this literature has evaluated investments in natural resources and flexible manufacturing (e.g., Brennan & Schwartz, 1985; Triantis & Hodder, 1990), analyzed the optimal timing of investing in land development (e.g., Titman, 1985), and studied the relationship between options to alter operating scale and the value of the firm (e.g., McDonald & Siegel, 1985; Pindyck, 1988; Majd & Pindyck, 1989). Pindyck (1991) and Dixit (1992) reviewed the literature on investment under uncertainty, and Dixit and Pindyck (1994) provided extensive discussions of theoretical advances. Two recent developments relating to strategy are noteworthy, however. First, research has paid increasing attention to the competitive environment surrounding firms' investments and the strategic aspects of real options, which have important implications for competitive strategy (e.g., Kulatilaka & Perotti, 1998; Grenadier, 2000; Smit & Trigeorgis, 2004). Second, research has also used real options theory to analyze investments in building strategic resources such as R&D, as well as other corporate development activities such as acquisitions and diversification, in the broader context of corporate strategy (e.g., Childs & Triantis, 1999; Matsusaka, 2001; Bernardo & Chowdhry, 2002; Pacheco-de-Almeida & Zemsky, 2003).

Compared to the large amount of theoretical work in this literature, there have been relatively few large-scale empirical studies, a point lamented by Schwartz and Trigeorgis (2001) and others. The available empirical analyses of real options in finance and economics have largely continued the focus of analytic work in the areas of natural resource investments and real estate development (e.g., Paddock, Siegel, & Smith, 1988; Quigg, 1993; Moel & Tufano, 2002), and have also examined the implications of particular options for the value of the firm (e.g., Berger, Ofek, & Swary, 1996). Empirical work on investing in strategic resources and corporate development is lacking, however, and option implementation issues related to organization, incentives, and the like have yet to be probed in more depth (Trigeorgis, 1996).

The Development of Real Options Theory in Strategic Management

Initial interest in real options in the field of strategic management began to emerge in the early 1980s, when management researchers first expressed dissatisfaction with traditional financial techniques such as the net present

value (NPV) approach to resource allocation and strategic decision making (e.g., Hayes & Garvin, 1982). These techniques make it hard to account for follow-on investment opportunities often embedded in a corporate investment project, or to capture managers' flexibility in adapting their decisions to evolving market and technological uncertainty, a view also shared by financial economists such as Myers (1984) and Kester (1984).

Kogut was among the first to formally conceptualize and empirically test real options in strategic management. His seminal work started in the context of multinational corporations (MNCs) and the coordination of their operations across countries. In a series of articles, Kogut (1983, 1985, 1989) maintained that multinational operations confer the MNC a string of real options in order to capitalize on the high levels of uncertainty and heterogeneous opportunities present across countries. For instance, he suggested that international investment confers the MNC valuable growth options, and an initial investment in a foreign country often carries a large option value, since the investment can unlock opportunities for future expansion. Kogut also emphasized that the MNC holds a portfolio of switching options that offer operating flexibility by allowing the firm to shift value chain activities across geographically dispersed subsidiaries as uncertain environmental conditions evolve.

A number of studies have expanded Kogut's initial contributions in several concrete ways. Kogut and Kulatilaka (1994a), for example, developed a model that captures the option value of production switching between two country locations in the presence of volatile exchange rates. Kogut and Chang (1996) empirically tested the idea that an initial investment may serve as a platform for subsequent expansion, and they found that Japanese firms' direct investments in the U.S. were triggered by appreciation of the Japanese yen. Miller and Reuer (1998a, 1998b) studied U.S. MNCs' economic exposures to foreign exchange rate movements, and they showed that firms with greater FDI have lower exposures, and that such exposures also tend to be asymmetric, which is consistent with the presence of real options. Allen and Pantzalis (1996) and Tang and Tikoo (1999) provided evidence that the stock market values the breadth of MNCs' international operations, supporting the notion of switching options available to the firms. More recently, Reuer and colleagues couched the benefits of operating flexibility in terms of the downside risk reduction from multinational investments (Reuer & Leiblein, 2000; Tong & Reuer, 2007), and they suggested and found that the extent to which MNCs can benefit from geographically dispersed operations is tempered by certain organizational factors that increase coordination and switching costs.

Kogut's pioneering contributions also pertained to the areas of governance and organizational choice in the corporate strategy domain. He provided the first theoretical arguments and empirical evidence that joint ventures (JVs) provide firms real options to expand sequentially into new and uncertain markets (Kogut, 1991). By investing in a JV, a firm is able to limit its downside losses to an initial, limited commitment, while also positioning itself to expand, but only if future conditions turn out favorably. In line with the theory, he found that the firm undertakes expansion by exercising the option by buying out its partners when the JV experiences a positive demand shock, but the firm continues to hold onto its investments in the JV when negative demand signals materialize.

A significant amount of theoretical and empirical research that followed has sought to extend this paper by examining the firm's choice of particular governance modes and related governance design issues. First, using formal models, Chi and colleagues have examined the circumstances under which the option to acquire or sell out a JV provides positive economic value for partners, investigated the conditions under which firms may hold the option rights, and analyzed governance structure issues such as the allocation of equity stakes between the partners (Chi & McGuire, 1996; Chi, 2000). Reuer and colleagues studied the real options embedded in various types of JVs (Reuer & Tong, 2005, 2007; Tong, Reuer, & Peng, 2008), and their findings indicated that JVs enhance firms' growth option values, yet only under some well-defined conditions. Second, Folta (1998) studied firms' decisions to undertake JVs versus acquisitions by viewing JVs as providing deferral options and sequential commitments, and he found that firms are more likely to invest in JVs over acquisitions when facing high levels of uncertainty. Folta and Miller (2002) built on Kogut's (1991) focus on option exercise decisions, but went beyond JVs to investigate minority equity investments. Building on Dixit and Pindyck (1994) and continuing Folta's (1998) focus on deferral options, Folta and colleagues examined firms' market entry decisions and presented findings consistent with real options theory (Miller & Folta, 2002; Folta & O'Brien, 2004; Folta, Johnson, & O'Brien, 2006). Collectively, this set of empirical evidence has begun to develop toward a real options theory of market entry and organizational governance that can complement existing theories: market entry modes differ in their attributes and embedded options, and they respond to uncertainty in different ways, leading firms to use them discriminately to structure their investments.

Around the same time as Kogut's work, Bowman and Hurry (1987, 1993) were working to develop an option theory based perspective of strategic management. Bowman and Hurry (1993) proposed options as a strategy

heuristic for understanding sequential resource commitments under uncertainty, and central to their theory development is the notion that the options lens “offers an economic logic for the behavioral process of incremental resource investment” (p. 760). Hurry, Miller, and Bowman (1992) found that Japanese venture capitalists tend to make small individual investments, yet a large number of investments, in order to capture a wide range of future opportunities, which they suggested is consistent with an ‘options strategy’ of seeking new technology. McGrath (1997) advanced a real options logic of technology options by suggesting that firms can make so-called amplifying preinvestments to influence uncertainty to their advantage; in a subsequent paper, she developed the notion that entrepreneurial initiatives can be viewed as real options and suggested that they be managed using real options reasoning (McGrath, 1999). In parallel to some of these lines of research, Kogut and Kulatilaka (1994b, 2001, 2003) aimed to integrate the literatures on real options and capabilities by proposing that real options theory provides a heuristic framing of viewing capabilities as generating platforms to respond to future uncertain opportunities.

Given the strategy field’s interest in understanding the actual behaviors of firms (Rumelt et al., 1994), it is not surprising that compared to real options research in finance and economics, research in strategy has paid considerably more attention to issues surrounding option implementation. While in principle real options theory can be applied to evaluate resources and strategic investments that are not publicly traded (Mason & Merton, 1985), strategy researchers have long suggested that various challenges can surround both the valuation and implementation (e.g., creation, maintenance, and exercise) of real options in organizations, in part due to several issues accompanying “domain translation” (Kogut & Kulatilaka, 2004). Indeed, this basic idea finds its roots in the initial contributions in the field and has run through the whole stream of real options research in strategy. For example, Kogut (1985) pointed to the difficulty that managers may have in recognizing valuable options embedded in the firm’s investments, a view also shared by Bowman and Hurry (1993). Moreover, just because a firm recognizes the embedded options does not mean that it has the management and organizational system to support their implementation (Kogut, 1989; Kogut & Kulatilaka, 1994a, 1994b). In addition, managers might not use the correct information to assess real options or might evaluate them incorrectly due to the lack of suitable proxies (Bowman & Moskowitz, 2001; Miller & Shapira, 2004). Finally, managerial and organizational factors might further alter option maintenance and exercise decisions: managers may be prone to escalation of commitment, they may not follow the optimal

exercise policies due to incentive problems, and they may find it hard to monitor the complex cues for exercise because of bounded rationality (Kogut, 1991; Garud, Kumaraswamy, & Nayyar, 1998; McGrath, 1999; Coff & Lavery, 2001; Adner & Levinthal, 2004).

The Importance of Real Options Theory for Strategic Management

Real options theory provides a set of analytic tools and heuristics to evaluate and deal with the uncertainty that pervades strategic decisions. Indeed, Rumelt et al. (1994, p. 26) identified uncertainty as among the top five “monkey wrenches” that inspired research departing from the neoclassical theory of the firm, and that has given rise to the birth of the strategic management field. Given the essential role of uncertainty in strategic decisions, we suggest that the increased importance of real options theory for strategic management can be explained by at least three factors that may also suggest why real options theory is unique.

First, real options theory requires research to revisit the received wisdom, and offers unique predictions, on firms’ decisions for many types of strategic choices under uncertainty. Consider the following three examples. As alluded to earlier, the real options view challenges the traditional perspective of joint ventures as marriages, under which longevity and stability were key indicators of success. According to real options theory, firms can unlock value at the joint venture termination stage, and an important role exists for joint ventures that are transitional investments by design. As a second illustration, foreign direct investment has long been considered a solution to the substantial transaction costs accompanying the market exchange of technology or other assets. By contrast, real options theory instead emphasizes dynamic efficiency gains, downside risk reduction, and the firm’s ability to seize upside opportunities over time by shifting value chain activities across borders in response to different uncertainties. Finally, at a more general level, real options theory provides new rules for resource investments by suggesting that real options shift firms’ investment thresholds away from the NPV0 criterion. While the details on the threshold effects of various real options have been illustrated elsewhere (Pindyck, 1988; Dixit & Pindyck, 1994; Trigeorgis, 1996), the insight offered by real options analysis can be briefly summarized as follows: a firm may use a reduced investment threshold and decide to invest even if the NPV is negative, if the embedded growth options are sufficiently valuable; by contrast, a firm may use an elevated investment threshold and decide not to invest even if the NPV is positive, if the

embedded deferral options are sufficiently valuable and the associated opportunity costs of investing in the current period are significant.

Second, real options theory uniquely posits an asymmetric payoff structure for investments with embedded options by suggesting that real options enable firms to reduce downside risk while accessing upside opportunities. The asymmetry in performance outcomes is due to the discretionary decision rights that options create, i.e., the right to select an outcome in the future only if it is favorable. Compared to other theories, real options theory therefore suggests that the greater the level of uncertainty, the higher the potential payoff to the option holder, given that the initial investment is limited and downside losses are contained (Bowman & Hurry, 1993; Hull, 2003). Another key aspect the theory emphasizes is that maintaining flexibility under uncertainty has option value, and this value can account for a substantial proportion of the value of many investments. Theory and empirical findings also suggest that such option value varies significantly across firms and industries, and of importance to strategic management is what the sources of heterogeneity might be and how option value influences firms' strategic choices and resource allocation policies (Kester, 1984; Tong & Reuer, 2006).

Third, real options theory sheds new light on firms' resource allocation processes by informing strategic decision making. Strategic planning has long embraced such concerns as follow-on opportunities, incremental resource commitments, and sequential management of information and uncertainty, which are all central to firm strategy; yet by their nature planning models lacked the kind of tight decision criteria prescribed by investment models in traditional finance theory. Real options theory can help improve strategic decision making by bringing the discipline of financial markets into qualitative strategic planning tools, and also by incorporating strategic realities into traditional capital budgeting models that do not explicitly account for the value of flexibility and managerial discretion (Trigeorgis, 1996; Amram & Kulatilaka, 1999). While effective implementation of real options analysis for resource allocation needs to overcome organizational and other challenges, real options theory holds out the promise of integrating strategic and financial analyses for corporate strategy (Bettis, 1983; Myers, 1984).

OVERVIEW OF THE VOLUME

The above three reasons why real options theory has become important for strategic management also correspond to three major streams of real

options research in strategic management, which we label as real options investment decisions, implementation of real options, and performance outcomes of real options, as summarized in Fig. 1. Below we outline the three streams of research, and we use this framework to structure the chapters in this volume and highlight their contributions.

Before discussing the individual chapters that make up this volume, it is fitting to describe the development of this collection as well as offer our thanks to several people and institutions, without whose support this project would not have been possible. In late 2004, we identified scholars doing research on the above three topics and invited them to contribute original research papers to a volume devoted to real options in the *Advances in Strategic Management* series. In June of 2006, roughly forty authors and participants gathered at the Kenan-Flagler Business School at the University of North Carolina for a conference intended to help the authors develop their papers as well as prompt discussion and debate on real options theory in strategy. Also for these purposes, we invited several scholars doing research in different streams within strategic management to serve as session facilitators to exchange ideas about the future of real options. We owe a special thanks to those who served as facilitators for the various sessions at the conference: Gautam Ahuja (University of Michigan), Connie Helfat (Dartmouth College), Don Lessard (MIT), and Arvids Ziedonis (University of Michigan). We also are grateful for the assistance provided by staff members and doctoral students at UNC throughout this project. Finally, we

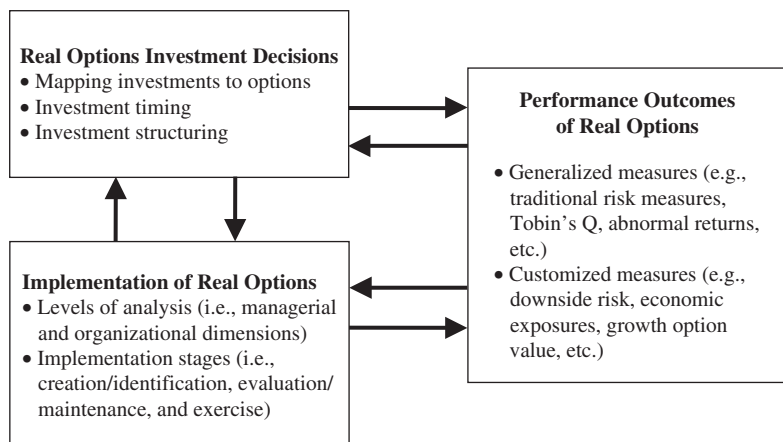


Fig. 1. A Framework for Real Options Research in Strategic Management.

would like to acknowledge the Center for Entrepreneurial Studies and the Kenan Institute for Private Enterprise at UNC that provided generous conference funding.

Advances in Real Options Research in Strategy

We begin this volume with a section including four chapters that delineate the recent advances in real options research in strategy, given that little work has systematically reviewed and analyzed existing contributions in the field. The first chapter by Li, James, Madhavan, and Mahoney reviews key applications of real options theory in strategic management and proposes several areas for future research. Their review suggests that real options theory provides unique insights into firms' investment under uncertainty; in particular, the theory has thrown new light on two topics of significant interest to strategy researchers: investment and divestment, and organization and governance. The review also indicates that real options embedded in strategic investments are valuable and have important performance implications for the firm. Their work concludes that real options theory has the potential to develop into an emerging, dominant conceptual lens in strategic management. The next chapter by Li provides a systematic analysis of the theoretical and empirical contributions of real options theory within international strategy. Her analysis builds on a framework that overlays three critical topics of research in international strategy (multinationality, market entry mode, and market entry timing) with three major approaches used in existing real options research in the field (real options modeling, real options reasoning, and empirical testing). She also outlines potential contributions that real options theory could make to two major streams of research in international strategy: research on transaction costs economics and research on internationalization theory. The third chapter, by Cuypers and Martin, focuses on real options theory's applications in research on joint ventures, a particular investment and governance mode that has drawn a substantial amount of attention in the strategy field. Their synthesis of the real options literature on JVs highlights real options theory's connections with several alternative theories on JVs, and they also examine how various options can affect a JV's development within and across different stages of the venture's life cycle. The final chapter in this section by Reuer and Tong categorizes and critiques the empirical research strategies that have been used to test real options theory in strategic management. Their research discusses studies that examine the timing and structuring of firm's investments, and their

particular focus is on studies that examine the performance implications of firms' real options investments. Their analysis suggests that considerable evidence has accumulated for real options theory, and they also indicate the need to pay attention to the costs associated with real options within distinct investment stages as well as across different stages.

The chapters in this section suggest that real options theory is well suited for studying strategic decision making under uncertainty in various investment contexts, and they also call for more theoretical and empirical work that can help advance the theory in several concrete ways. In particular, there is a need to better articulate real options theory's link to other theories in the field and to specify the theory's appropriate boundaries. In addition, more and stronger tests are also required to fill the gap that still exists between theory and practice as well as to resolve some empirical inconsistencies documented in the literature. To better understand real options theory's applicability, researchers can extend the theory to new application areas, study several types of options that have received relatively less attention as well as option interactions, and pay more attention to the implementation aspect of real options.

Real Options and Strategic Investment Decisions

The second section focuses on firms' strategic investment decisions using real options theory. Research in this stream often starts by identifying different types of real options embedded in strategic investments. This research then examines how the presence or absence of these options may affect the timing and structuring of such investments under uncertainty and other environmental conditions. Research on the timing of investments has developed models to derive the optimal conditions under which firms are making investments (e.g., Kulatilaka & Perotti, 1998; Leiblein & Ziedonis, this volume; Lin & Kulatilaka, this volume) and has empirically tested whether the actual investment timing is consistent with real options theory's predictions (e.g., Kogut, 1991; Campa, 1994; Folta & O'Brien, 2004; Folta & O'Brien, this volume; Nerkar, Paruchuri, & Khair, this volume).

The chapters in this section contribute to existing research on the timing of investments in several ways. The first chapter by Lin and Kulatilaka extends previous theoretical research by considering firms' investment decision in a specific industry setting, network industries, where strategic advantages arising from early commitment generate a valuable strategic

growth option. Their study suggests that under high uncertainty, the strategic growth option often dominates the deferral option, thus reducing firms' investment thresholds and encouraging investments; in addition, the intensity of network effects enhances the value of the strategic growth option. The chapter by Folta and O'Brien examines the likelihood of firms making acquisition investments, which have embedded growth options and deferral options. They use a novel technique to isolate real options' effects on firms' investment thresholds, and they find that firms' thresholds affect the likelihood of acquisition in ways consistent with the theory's predictions. The chapter by Leiblein and Ziedonis applies real options theory to study firms' technological adoption strategies when there are multiple generations of technologies that are introduced successively. Their conceptual model identifies several conditions that differentially affect the value of deferral and growth options embedded in technological adoption, which in turn determines firms' optimal adoption strategy under those conditions. The chapter by Nerkar, Paruchuri, and Khaire extends recent research that views patents as real options, and they suggest that patents provide their holders with the right but not the obligation to sue potential infringers. They study the exercise of the option to sue in a novel setting – business method patents – and their findings suggest that the likelihood of a patent being litigated is positively associated with the value of the patent and the extent of disclosure in the patent.

Research on the structuring of investment has tended to focus on how firms structure their investments, such as the design of investment patterns and investment portfolios (e.g., Kogut, 1983; Hurry et al., 1992; Vassolo, Anand, & Folta, 2004; Anand, Oriani, & Vassolo, this volume). Research in this stream has also examined organizational governance and investment mode choice, such as alliances versus acquisitions (e.g., Chi & McGuire, 1996; Folta, 1998), assuming that a broader corporate investment decision is in place. The chapter by Anand, Oriani, and Vassolo in this section analyzes several factors that determine the value of a portfolio of real options and therefore can affect the composition of an option portfolio. Their core idea is that building an effective option portfolio requires attention to balancing growth and switching options, and they discuss how the value of an option portfolio depends on the width of the portfolio as well as the correlation among the underlying assets for each option. Their research thus also has useful implications for the implementation of real options, which is the focus of the four chapters in the next section.

Organizational and Managerial Dimensions of Real Options

Researchers have moved beyond strategic investment decisions to examine the implementation of real options in real organizations. While as a theory of investment, real options theory does not speak directly to managerial and organizational capabilities required for implementation, more research in this area can help to specify the theory's boundaries and enhance its managerial relevance. Most of the existing research in this area is conceptual in nature, describing various opportunities and challenges facing firms implementing real options. While specific topics vary, this research has tended to emphasize the importance of managerial or organizational dimensions during the various stages of option implementation, such as option creation and identification, option evaluation and maintenance, and option exercise.

Managerial and organizational factors can affect option implementation at different investment stages. For example, management processes and organizational structures can influence firms' identification of real options and their investments in real options (e.g., Kogut, 1985; Bowman & Hurry, 1993). The first chapter in this section, by Maritan and Alessandri, uses a capabilities perspective to link investments in real options to firms' resource allocation process. They first identify four components of the returns to an investment, deriving from industry-specific elements, as well as option and non-option elements, and they link these components to specific levers of the resource allocation process. They also suggest that research focus on the organizational and managerial aspects of the investment process from option creation to option exercise. This suggestion is consistent with the broader view that the evaluation, maintenance, and exercise of real options may need to deal with various management and organizational challenges (e.g., Kogut & Kulatilaka, 1994a, 1994b; Coff & Laverly, 2001; Adner & Levinthal, 2004). The following two chapters in this section further extend this view. Coff and Laverly suggest that managing real options in different organizational forms can incur different organizational costs, and therefore the organizational form that an option takes can have a profound effect on option exercise decisions. Their research also prescribes several organizational and management processes that may facilitate the management of real options in organizations and thus help to achieve real options theory's promise in strategic management. Adner recasts recent discussions on the appropriate applicability of real options theory to strategic management in terms of the characteristics of the resource reallocation process in organizations. His research considers some managerial and organizational

drivers of mismatches between initial resource allocation logics and subsequent resource reallocation realities, and it highlights the need for a better understanding of the resource reallocation process in order to improve the appropriate usage of real options logic in organizations. In contrast to these chapters focusing on the challenges surrounding option implementation at different stages, the final chapter in this section, by Fister and Seth, analyzes one specific management challenge – how to encourage employees' investment in firm-specific human capital – using real options theory. Their application of real options theory points to several conditions that would lead to the use of certain contractual mechanisms to encourage such investment, and they discuss how various mechanisms might serve such a purpose through their impact on the value of the various options embedded in employment relationships.

Performance Implications of Real Options

The final section of the volume relates to an emerging stream of research that empirically investigates the performance implications of real options. As observed in Reuer and Tong (this volume), research within this stream has used both so-called generalized measures and customized measures to study the firm outcomes of real options investments. Generalized measures refer to market returns, market values, traditional risk measures, as well as other proxies that have also been used for testing other theories. Customized measures, in contrast, are specifically geared toward testing the unique payoff structure associated with particular real options, and the existing research has used such measures as downside risk, growth option value, abandonment option value, asymmetric exposures to uncertainties, and so forth.

The first chapter in this section by Chi and Levitas conceptualizes patents as technology options and empirically examines the option value of a firm's patent portfolios. Their research isolates the real options' effects by considering factors that tend to influence option value but not cash flow value. They do so by investigating how flexibility in excising options embedded in patents (proxied by citation dispersion) may moderate the effect of patent citations on the firm's market value, based on a theorem that is developed in Merton (1973) and also discussed by Bowman and Hurry (1993). The findings show that patent citations have a more positive influence on firm value when the citations are more dispersed and when there is a higher level of uncertainty, both of which are consistent with real options theory. The

chapter by Oriani examines the value of a specific real option, i.e., technology switching option, which allows a firm to exchange an existing technology with a new technology. Specifically, he develops a model of the market value of the firm that explicitly incorporates a technology switching option, and he empirically tests the impact of this option on firms' value. His findings suggest that the technology switching option is valuable and that its value is enhanced for firms having a higher probability to exercise the option. The following chapter, by Alessandri, Lander, and Bettis, also empirically values specific real options, in this case corporate growth options. Their research builds on Kester's (1984) initial contribution to estimate a firms' value of growth options, using different valuation models that represent different assumptions and techniques. Their findings indicate that a firm's growth option value is a function of the macroeconomic and industry environment in which the firm operates, as well as firm-specific factors, suggesting the need for finer-grained study of real options at different levels of analyses. The final chapter in this section, by Guler, takes a different, yet complementary approach to studying the performance implications of real options. Her research investigates venture capitalists' investment policies in managing their portfolio companies, which are considered real options investments. Her findings indicate that firms differ in their capabilities to manage unsuccessful projects but not successful projects, reflecting earlier research's suggestion to focus on firm heterogeneity in studying the performance implications of real options.

FUNDAMENTAL QUESTIONS FOR REAL OPTIONS RESEARCH IN STRATEGIC MANAGEMENT

The set of chapters in this volume, combined with previous research, illustrate the increasing interest in real options theory in the strategy field. This work also demonstrates the rich theoretical content and wide empirical application of real options theory within strategic management. At the same time, real options theory is still at a relatively early stage of development, and many important issues will need to be tackled if the theory is to attain a status comparable to a number of other perspectives in currency in strategy research.

As an initial step in identifying and cataloguing some of the most pressing areas for research, we asked all of the participants at the conference to flag one or two key issues worthy of future research. Rather than simply listing

these research topics, we attempted to distill them into a smaller number of fundamental questions that scholars need to address. During this process, we were mindful of Rumelt et al.'s (1994) proposal of four fundamental questions in strategy to differentiate this field of inquiry. We offer these questions to highlight some issues that are fundamental to real options research in strategy, in order for the theory to obtain the promise that researchers have envisioned for it (Bowman & Hurry, 1993; Kogut & Kulatilaka, 2001; Barney, 2002; Mahoney, 2005). As fundamental questions, they may challenge the current state of knowledge, yet we also believe that efforts to work on these questions can not only help real options theory make greater contributions to the strategic management field, but they can also bring into focus the distinctive contributions that strategy research can make to real options theory. Below we discuss the four questions and we include some additional questions under each broader category.

How can Real Options Theory Address the Foundations of Strategy?

The field of strategic management is concerned with the firm's strategic choices and directions, which can have an enormous influence on organizational performance. In its most strict form, real options theory can help parameterize sources of uncertainty and attach values to the various options embedded in the firm's strategic decisions and investment choices. At a broader level and used in metaphoric terms, real options theory can offer a more positive view of uncertainty and a more constructive view of managerial discretion by advising firms to attend to key value drivers for the various options embedded in strategic choices.

Strategy is also fundamentally interested in the heterogeneity in firm behaviors and performance outcomes. Like some other theories within organizational economics, real options theory does not seek to address firm heterogeneity directly. However, there are significant opportunities to enhance the conversation between real options theory and firm heterogeneity. For example, firms must engage in strategic investments to build or acquire resources and capabilities under considerable uncertainty and ambiguity (Lippman & Rumelt, 1982; Barney, 1986; Dierickx & Cool, 1989). As one fundamental theory to aid in investment decision making (Dixit & Pindyck, 1994), real options theory can enhance our understanding of why firms may differ by modeling firm's investments in a more analytic way (e.g., Pacheco-de-Almeida & Zemsky, 2003). Indeed, research has suggested that firms can use real options theory to guide their investments to build

heterogeneous resources and capabilities (Baldwin & Clark, 1992, 2000). Firm heterogeneity can also be used to explain particular predictions of real options theory (Tong & Reuer, 2006). For instance, firm heterogeneity and the associated asymmetric expectations across firms may lie at the heart of the reason why firms may exhibit different investment behaviors when facing the same uncertainty, whether the investments are made for strategic factors, real options, or financial securities (Barney, 1986; Chi & McGuire, 1996; Hull, 2003).

The following subquestions connect to this broader question, which is of particular interest to strategic management:

- How can real options theory explain competitive advantage? How can firms facing the same options achieve differential performance? If firms face different options and this creates heterogeneity in resources and performance, what is the source of the firm heterogeneity in the first place?
- How can real options theory speak to corporate strategy? What are the implications of viewing the firm as a portfolio of options rather than a bundle of resources and capabilities? How can research use real options theory to explicate the development of core competencies?
- How important are the option properties of organizational governance structures in explaining boundary of the firm choices?

How can Real Options Theory Connect to Other Theories in Strategy?

The distinctiveness of real options theory for strategic investment decisions compared to other theories has been discussed in a previous section, and its uniqueness also explains strategy researchers' initial enthusiasm for the theory. Real options research clearly should continue to highlight the theory's unique aspects as a standalone theory, by further examining the critical questions and predictions the theory poses and by emphasizing the theory's unique constructs as well as the links between these constructs and strategic decisions or organizational performance. Equally important, conceptual and empirical research should work to tease apart the theory's predictions from those of rival theories and better identify real option theory's boundaries in strategic management.

Another way to advance strategy research on real options is to combine the theory with other theories to examine particular questions or phenomena. Such an integrated approach has the potential advantage of offering a more complete understanding of the questions examined, and recent research has made headway in this direction. For example, strategy research

has combined real options theory and transaction cost economics to explicate the conditions under which firms may invest in JVs and has considered related alliance design issues (e.g., Chi & McGuire, 1996). Connecting real options analysis with the resource-based view has the potential to improve the analysis of firms' corporate development trajectories such as the directions and patterns of diversification (e.g., Kim & Kogut, 1996; Matsusaka, 2001; Bernardo & Chowdhry, 2002). Incorporating the resource-based view and related notions of firm heterogeneity also holds the potential to explain the heterogeneous expectations and investment behaviors of firms facing the same external uncertainty (e.g., Tong & Reuer, 2006). In addition, integrating real options theory with other theories of organizational governance can better inform corporate strategy decisions such as make or buy decisions and the firm's vertical boundaries (e.g., Leiblein & Miller, 2003). Ample opportunities also exist for extending real options theory into contexts with agency problems, competitive rivalry, or other sources of endogenous uncertainty, and strategy research has barely started to consider real options theory in tandem with agency theory, game theory, or organizational learning theory.

Clearly, both of the two approaches are valuable and can be appropriate in different research designs, yet they also present some additional questions for scholars to consider when framing their research and designing studies on real options:

- In general, what is the best way forward for real options theory to make greater contributions to strategy research, as a standalone theory or as a theory integrated with others?
- More specifically, how should research implement each of the two approaches? What are some of characteristics of the questions to which real options theory should be applied? What phenomena can real options theory potentially explain better than other theories? What are the most interesting empirical horse races to be run?
- What other theories can be profitably combined with real options theory, and in what contexts? How can the boundaries between the theories be delineated carefully while also acknowledging shared concerns and points of connection?

How Important is Formalism in Real Options Theory?

Real options theory has been applied in strategy research in many different ways, yet in a certain sense, studies using the theory might be arrayed along a continuum ranging from very formal work to metaphoric applications. At

one extreme, formalism can help identify the workings of real options, isolate the embedded options, and pin down the option value drivers, yet important strategic realities may need to be assumed away or differences in assumptions in financial and strategic domains glossed over. At the other extreme, metaphoric usage of real options might better attend to strategic or organizational realities and broaden the applications to which option theory might speak, yet it may not reflect concerns or variables directly featured in formal models of real options. Therefore, the question is: how important is formalism in the use of real options theory in strategic management? In this connection, it is worth noting that such a question is useful to ask not just for a particular theory such as real options, but the question also has great applicability to other theories within the strategy field. The advancement of a good theory often needs to attend to such conflicting considerations.

Formalism need not be equivalent to an exclusive focus on option valuation models *per se*, however. For strategy research, what appears to be more important is to determine what questions to investigate that are core and interesting to our field of inquiry, whether using analytic or empirical models to evaluate investments or strategic choices in a rigorous fashion. Metaphorical usage of real options also can be valuable for certain applications. For example, research has suggested that corporate managers more often use real options theory as a framing device or decision framework rather than as a formal valuation tool, yet such practices have been powerful and have also in many ways transformed managers' and investors' views on strategic investments such as R&D, information technology, and other platform investments (Triantis, 2005). Clearly, the challenge for research is to determine the proper balance between formalism and metaphoric usage of the theory so that applications of real options are still sound and sensitive to alternative explanations. While research needs to consider the kind of topics and contexts for real options as well as weigh the level of maturity achieved and knowledge accumulated, the following subsidiary questions also might be considered:

- When should real options theory be used as an analytic tool versus a heuristic? How can research establish a strong connection to the features in formal models in using real options theory as a heuristic?
- How can strategic and organizational realities be incorporated into analytical models of real options? How can real options be modeled more explicitly and precisely in order to link theory with empirics more tightly?
- What is the best way to determine the real options characteristics in various assets and investments? When does it pay to designate a strategic choice or investment as an option?

What is the Role of Management and Organization in Real Options Theory?

To the extent that the trading of options on financial assets with posted prices may still be subject to inefficiencies, great frictions must exist in the acquisition of real options on strategic assets in factor markets as well as in the development of such strategic options within organizations. For example, organizations are run by boundedly rational managers with their own cognitive limitations and behavioral biases, and they may have limited attention and may face difficulties in recognizing complex cues accompanying multiple sources of uncertainty (Kogut, 1991; Miller & Shapira, 2004; Barnett, 2005). In addition, organizations may not have the appropriate structures or supportive systems in place (Kogut, 1985; Coff & Lavery, 2001), and managers might also misuse their discretion and deviate from optimal decision criteria as a consequence (Trigeorgis, 1996; McGrath, Ferrier, & Mendelow, 2004). Real options theory is a theory of investment decision making that places a high demand on managerial and organizational capabilities for execution. Despite the soundness of the theory, any of the factors above can threaten to destroy option value at various stages of option implementation (e.g., recognition, creation, maintenance, exercise, etc.).

Viewed from another perspective, however, to the extent that managers and firms factor these considerations into their decision calculus and develop corrective mechanisms, they can help contribute to option value creation. Indeed, the existence of frictions in the process of the development and exchange of real options has helped open up an important opportunity for strategic management research to make significant contributions to real options theory. One of strategic management's distinctive competences is to provide a holistic view of the firm by bridging strategy formulation to management and organization. Employing this distinctive competence, strategic management research can tackle important questions at the heart of the successful implementation of real options, and such work will advance both the theory as well as its practice, which research in other fields will find hard to achieve (e.g., Hartmann & Hassan, 2006). Toward this end, the following subquestions are put forth:

- How do specific management and organizational factors matter for real options? How does the importance of management and organization vary across firms and strategic contexts?
- How do firms actually implement real options analysis? How do managers perceive various sources of uncertainty and apply real options analysis or reasoning in *ex ante* strategic decision making processes?

- What challenges in other option investment stages (e.g., recognition, creation, and maintenance) should we devote more attention to, in addition to challenges surrounding option exercise? How should firms measure and reward the creation, maintenance, and exercise of options in organizations, especially when some managers develop assets and others operate them?

Clearly, the four questions are not isolated, and they connect to one another to address some common themes or broader concerns. For example, the first two questions primarily focus on how real options theory can better contribute to strategy research, and the latter two turn to ask how strategic management can make significant contributions to real options theory. Also, these questions are likely not collectively exhaustive, and other useful questions can certainly be suggested; nevertheless, we believe that most of them can be related to the four questions here in one way or another. These four questions are therefore fundamental to our understanding of real options theory in strategic management and key to future progress of research in this area.

This volume illustrates how real options theory has significantly contributed to strategic management research, as well as how scholars in strategic management are uniquely positioned to advance the theory. The chapters demonstrate the diverse applications of real options theory in strategy, and they also point to a range of methodologies and analytical lenses that future strategy research could leverage to improve existing understanding of the theory. The volume and the conference have also identified several pathways for real options research to advance and develop into a major theoretical perspective in the field. Our hope is that this volume can serve as a useful guide to real options research in strategic management for interested readers, and as a catalyst for additional research on this theory in coming years.

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